1 Time-dependent perturbation theory

We now turn to time-dependent perturbation theory.[1] As a taste of what is to come, we will use this to compute cross sections and decay rates: these are the main observables in a lot of atomic, nuclear, and particle physics.

Setup

Let us start with the Schrödinger equation[2]:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle , \tag{1}$$

with the Hamiltonian split into a free part $\hat{H}_0$ and an interaction $\hat{V}(t)$:

$$\hat{H} = \hat{H}_0 + \hat{V}(t). \tag{2}$$

The free part describes the evolution in the absence of interactions (we assume no time dependence in H_0), while $\hat{V}(t)$ encodes the effect of some interaction potential (now with possible time-dependence). We will treat $\hat{V}(t)$ as a small perturbation and expand the solution to the Schrödinger equation in powers of $\hat{V}(t)$.

In the absence of the interaction, the time evolution of some state $|\psi(t)\rangle$ is simply

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}_0 t} |\psi(0)\rangle . \tag{3}$$

Let us decompose $|\psi\rangle$ in the eigenstates of $\hat{H}_0$, labelled $|n\rangle$ with $\hat{H}_0 |n\rangle = E_n |n\rangle$. Their eigenfunctions $\phi_n(\vec{x}) = \langle \vec{x}|n\rangle$ form a complete orthonormal basis, $\langle m|n\rangle = \delta_{mn}$ and $\sum_n |n\rangle \langle n| = \mathbb{1}$. We write

$$|\psi\rangle = \sum_n c_n |n\rangle , \qquad \sum_n |c_n|^2 = 1, \tag{4}$$

so that $c_n = \langle n|\psi\rangle$. Acting the free time-evolution operator on $|\psi\rangle$, we get

$$|\psi(t)\rangle = \sum_n c_n\, e^{-\frac{i}{\hbar}E_n t} |n\rangle , \tag{5}$$

using the fact that $e^{-\frac{i}{\hbar}\hat{H}_0 t} |n\rangle = e^{-\frac{i}{\hbar}E_n t} |n\rangle$.

In the absence of perturbations, what is the probability that the state $|\psi(t)\rangle$ remains unchanged after some time t? This is given by the squared modulus of the autocorrelation function:

$$P_\psi(t) = |\langle\psi|\psi(t)\rangle|^2 = \left|\sum_n |c_n|^2\, e^{-\frac{i}{\hbar}E_n t}\right|^2 . \tag{6}$$

If at $t = 0$ the state is an eigenstate of $\hat{H}_0$, say $|n_0\rangle$, then $c_n = \delta_{nn_0}$ and $P_\psi(t) = 1$ for all t: the state does not change. However, if $|\psi\rangle$ is a superposition of different eigenstates, the probability oscillates in time.

When $|\psi\rangle$ is a superposition of multiple eigenstates, what is the long-time ($t \to \infty$) average of $P_\psi(t)$? Consider what happens to the cross terms after squaring the sum in Eq. (6).

[1] These notes adapted from the particle physics course.

[2] Recall that the ket $|\psi(t)\rangle$ gives the wavefunction via $\psi(\vec{x}, t) = \langle\vec{x}|\psi(t)\rangle$. To ensure probability conservation, $\hat{H}$ must be Hermitian, $\hat{H} = \hat{H}^\dagger$, and the state must be normalized, $\langle\psi|\psi\rangle = 1$. Note that the abstract ket depends only on time, while the wavefunction carries dependence on both time and space; hence the Schrödinger equation for $\psi(\vec{x}, t)$ involves $\partial/\partial t$, whereas the one for $|\psi(t)\rangle$ involves d/dt.

Take the case of a two-state system, $|\psi\rangle = c_1 |1\rangle + c_2 |2\rangle$. Parameterize the coefficients as $c_1 = \cos\theta$ and $c_2 = \sin\theta$ (using $|c_1|^2 + |c_2|^2 = 1$). What is $P_\psi(t)$? Express your answer using only sine functions, using $\sin^2(\theta/2) = (1 - \cos\theta)/2$. These exercises illustrate why oscillation phenomena arise in quantum mechanics. For example, the fact that neutrinos oscillate means they are not produced as energy eigenstates of the Hamiltonian.

Turning on the interaction

Now let us consider $\hat{V} \neq 0$. One can take the following general form for the time-evolved solutions to Schrödinger's equation:

$$|\psi(t)\rangle = \sum_n c_n(t)\, e^{-\frac{i}{\hbar}E_n t} |n\rangle\,, \tag{7}$$

exactly as in Eq. (5), except that the coefficients $c_n(t)$ are now time-dependent. As we will see, the interaction can cause transitions between eigenstates of $\hat{H}_0$, and so the expansion coefficients must evolve. This is equivalent to saying that the interactions mix the free eigenstates: the true energy eigenstates of $\hat{H}$ are different from those of $\hat{H}_0$.

Plugging Eq. (7) into the Schrödinger equation, and assuming $\hat{V}$ is time-independent,[3] we obtain

$$i\hbar \sum_k \dot{c}_k(t)\, e^{-\frac{i}{\hbar}E_k t} |k\rangle = \hat{V} \sum_k c_k(t)\, e^{-\frac{i}{\hbar}E_k t} |k\rangle\,, \tag{8}$$

where the free-evolution terms $E_k c_k$ cancel on both sides. Taking the inner product with $\langle n| e^{+\frac{i}{\hbar}E_n t}$, we get

$$\boxed{i\hbar\, \dot{c}_n(t) = \sum_k c_k(t)\, e^{i\omega_{nk}t}\, V_{nk},} \tag{9}$$

where we use our standard notation

$$\omega_{nk} = \frac{E_n - E_k}{\hbar}, \qquad V_{nk} = \langle n| \hat{V} |k\rangle\,. \tag{10}$$

(Caution: V_{nk} is the matrix element of $\hat{V}$ in the Schrödinger picture.) This is a set of coupled first-order differential equations for the coefficients $c_n(t)$ and is entirely equivalent to the Schrödinger equation.

Let us intentionally fail to solve this system, and see why perturbation theory is the right approach. If we integrate both sides, we get

$$c_n(t) = c_n(0) - \frac{i}{\hbar} \sum_k \int_0^t dt'\, c_k(t')\, e^{i\omega_{nk}t'}\, V_{nk}. \tag{11}$$

This is not a solution: the right-hand side still involves $c_k(t')$. However, inserting this expression back into the right-hand side yields an infinite series of nested integrals, with each insertion bringing an extra factor of V_{nk}. If $\hat{V}$ is small (in the appropriate sense), each successive term is suppressed relative to the previous one. This is the central idea of perturbation theory: we truncate the series at some finite order. The procedure of inserting the solution back into itself is what gives rise to the Dyson series.

[3] You can also think of a potential that was 0 for $t < 0$ and then suddenly turned on at $t = 0$.

Perturbative expansion

We expand the coefficients in powers of the perturbation:

$$c_n(t) = c_n^{(0)} + c_n^{(1)}(t) + c_n^{(2)}(t) + \cdots, \tag{12}$$

where $c_n^{(j)}$ collects all contributions of order $\hat{V}^j$ (i.e., involving j insertions of the interaction). By construction, each order depends only on the solutions at lower orders, so the system can be solved iteratively.

We start with an initial state $|\psi(0)\rangle = |i\rangle$, meaning $c_n(0) = \delta_{ni}$. At zeroth order ($\hat{V} = 0$), the state does not evolve away from $|i\rangle$:

$$c_n^{(0)} = \delta_{ni}. \tag{13}$$

First order. Plugging $c_k^{(0)} = \delta_{ki}$ into Eq. (9), we get

$$i\hbar\, \dot{c}_f^{(1)}(t) = e^{i\omega_{fi}t}\, V_{fi}, \tag{14}$$

which, with the initial condition $c_f^{(1)}(0) = 0$ for $f \neq i$, integrates to

$$c_f^{(1)}(t) = -\frac{i}{\hbar}\int_0^t dt'\, e^{i\omega_{fi}t'}\, V_{fi} = -V_{fi}\,\frac{e^{i\omega_{fi}t} - 1}{\hbar\omega_{fi}} = -V_{fi}\,\frac{e^{i\omega_{fi}t} - 1}{E_f - E_i}. \tag{15}$$

The transition probability to the state $|f\rangle$ is therefore

$$P_{i\to f}(t) = |c_f^{(1)}(t)|^2 = |V_{fi}|^2\,\frac{\sin^2\!\left(\frac{\omega_{fi}\,t}{2}\right)}{\left(\frac{\hbar\omega_{fi}}{2}\right)^2}. \tag{16}$$

In the long-time limit ($t\,|\omega_{fi}| \gg 1$), we use the representation

$$\lim_{t\to\infty}\frac{\sin^2(\omega t/2)}{(\omega/2)^2} = 2\pi t\,\delta(\omega), \tag{17}$$

which, using $\delta(\omega_{fi}) = \hbar\,\delta(E_f - E_i)$, gives

$$P_{i\to f}(t) \xrightarrow{t\to\infty} \frac{2\pi t}{\hbar}\,|V_{fi}|^2\,\delta(E_f - E_i). \tag{18}$$

The transition *rate* is then

$$\boxed{\Gamma_{i\to f} = \frac{P_{i\to f}}{t} = \frac{2\pi}{\hbar}\,|V_{fi}|^2\,\delta(E_f - E_i).} \tag{19}$$

This is Fermi's Golden Rule in its simplest form. The rate is proportional to the squared matrix element $|V_{fi}|^2$ and the delta function enforces energy conservation, $E_f = E_i$.

The case $f = i$: One can ask what happens if $f = i$, i.e., the transition is back to the initial state. In this case, the first-order correction to the amplitude is

$$P_{i\to i}(t) = |c_i^{(0)} + c_i^{(1)}(t)|^2 = |1 - \frac{i}{\hbar}\int_0^t dt'\, V_{ii}|^2 = 1 + |V_{ii}|^2\, \frac{t^2}{\hbar^2}. \tag{20}$$

Note that the Hermiticity of the Hamiltonian required V_{ii} to be real, so we only get a contribution at higher order. (With an imaginary part, the probability of remaining in the initial state decreases with time at a rate of $\Gamma_{i\to i} = -\frac{2}{\hbar}\,\mathrm{Im}(V_{ii})$, which is precisely the term of $\mathcal{O}(t)$ we dropped above.) This idea of a non-Hermitian effective Hamiltonian with an imaginary part encoding decay widths is widely used in particle physics, nuclear physics, and quantum optics, but it does not tell you about to what states the system is decaying into, and thus is not a description of the dynamics. Note also that if V_{ii} is purely real, then $\Gamma_{i\to i} = 1$: the probability of remaining in the initial state is 1 at all times, except for higher order corrections. This piece is often called the "forward scattering" contribution, and it is important to include it in order to preserve unitarity (probability conservation) at each order in perturbation theory.

If you inspect the $i \to i$ coefficients at all orders, you will find that they sum up to a pure phase factor $e^{-i\delta_i}$, which does not affect the probability of remaining in the initial state:

$$c_i(t) = 1 - \frac{i}{\hbar}V_{ii}t - \frac{V_{ii}^2 t^2}{2\hbar^2} + \cdots = e^{-i\frac{V_{ii}t}{\hbar}}, \tag{21}$$

where it is said that we "resummed" all orders to get the exact exponential form (a.k.a. as exponentiation of the forward scattering amplitude). Now, let us go back to the state evolved in time and inspect its i-th component:

$$|\psi(t)\rangle = \sum_n c_n(t)\, e^{-\frac{i}{\hbar}E_n t}\,|n\rangle \quad \text{contains} \quad c_i(t)\, e^{-\frac{i}{\hbar}E_i t}\,|i\rangle = e^{-i(V_{ii}+E_i/\hbar)t}\,|i\rangle\,. \tag{22}$$

This shows that the effect of the diagonal transition is to shift the energy of the initial state by V_{ii}, which is called the *phase shift* or *energy shift*. So, while the absolute phase of a state is unphysical, the energy shift is a physical observable. By the way, note that this result is precisely the time-independent perturbation theory result for the energy shift of the state $|i\rangle$: this is embedded in the time-dependent formalism, as expected.

Second-order corrections

We now use the first-order solution to compute the second-order correction $c_f^{(2)}(t)$. From Eq. (9), keeping contributions at second order in $\hat{V}$:

$$i\hbar\, \dot{c}_f^{(2)}(t) = e^{i\omega_{fi}t}\, V_{fi}\, c_i^{(1)}(t) + \sum_{k\neq i} e^{i\omega_{fk}t}\, V_{fk}\, c_k^{(1)}(t), \tag{23}$$

where the first term isolates the $k = i$ contribution (since $c_i^{(1)}$ obeys a slightly different equation). Inserting Eq. (15), we find that, after integration and passage to the long-time limit, the result can be written in a compact form:

$$P_{i\to f}(t) = |c_f(t)|^2 = |\mathcal{M}_{fi}|^2 \cdot \frac{2\pi t}{\hbar}\,\delta(E_f - E_i), \tag{24}$$

where we have defined the *transition amplitude* (to second order)

$$\mathcal{M}_{fi} = V_{fi} + \sum_k \frac{V_{fk}\,V_{ki}}{E_i - E_k} + \cdots \tag{25}$$

The pattern is clear: at each successive order, one inserts a complete set of intermediate states $|k\rangle$, weighted by the energy denominator $1/(E_i - E_k)$ and an additional matrix element of $\hat{V}$. The first term describes a direct transition $|i\rangle \to |f\rangle$; the second describes a transition through an intermediate (virtual) state $|k\rangle$, and so on.

In full generality, at any order in perturbation theory, the transition rate is given by

$$\boxed{\Gamma_{i\to f} = \frac{2\pi}{\hbar}\,|\mathcal{M}_{fi}|^2\,\delta(E_f - E_i), \qquad \mathcal{M}_{fi} = V_{fi} + \sum_k \frac{V_{fk}\,V_{ki}}{E_i - E_k} + \sum_{k,\ell} \frac{V_{f\ell}\,V_{\ell k}\,V_{ki}}{(E_i - E_\ell)(E_i - E_k)} + \cdots} \tag{26}$$

This is Fermi's Golden Rule to all orders. The delta function enforces energy conservation between the initial and final states, but the intermediate states $|k\rangle$, $|\ell\rangle$, ... need *not* conserve energy. These are called *virtual* states.

Each term in $\mathcal{M}_{fi}$ has a natural diagrammatic representation. The first-order term V_{fi} corresponds to a single vertex connecting $|i\rangle$ to $|f\rangle$. The second-order term involves two vertices connected by an intermediate state $|k\rangle$, which propagates with the factor

$$\text{“Propagator”} = \frac{1}{E_i - E_k}. \tag{27}$$

Feynman diagrams are not far from this picture: they provide a systematic, pictorial bookkeeping of all the terms in $\mathcal{M}_{fi}$. We will see this structure appear again when we discuss the Dyson series for the time-evolution operator and the propagator.

> You might ask: should energy not be conserved at each vertex, requiring $E_k = E_i$? The answer in non-relativistic quantum mechanics is no. The intermediate virtual state $|k\rangle$ is not directly observed. At each vertex, *momentum* is conserved, but not energy. The degenerate intermediate state with $E_k = E_i$ requires special treatment.

Old-fashioned perturbation theory and covariance

At this point, we can glimpse something rather remarkable that happens in quantum field theory. One can recover Lorentz covariance by including virtual states that propagate not only forwards, but also *backwards* in time. Through the Feynman–Stückelberg interpretation, a particle propagating backwards in time is reinterpreted as its antiparticle propagating forwards in time.[4] In non-relativistic quantum mechanics, manually including such contributions is called “old-fashioned perturbation theory,” but in quantum field theory, it is mandatory to preserve covariance and causality.

Consider the second-order amplitude. The standard (forward-propagating) diagram contributes the propagator $1/(E_k - E_i)$, where E_k is the energy of the single intermediate particle. Now add a diagram where the intermediate state goes backwards in time. In this diagram, the intermediate state comprises three particles (the original, plus a particle–antiparticle pair), so $E_{\text{virtual}} = E_i + E_k + E_f$. Since the energy-conservation delta function will eventually enforce $E_f = E_i$, this second propagator becomes

$$\frac{1}{E_{\text{virtual}} - E_i} = \frac{1}{E_i + E_k + E_f - E_i} = \frac{1}{E_i + E_k}. \tag{28}$$

Summing the two diagrams (the amplitudes are otherwise identical), we obtain

$$\mathcal{M}^{(2)}_{fi} = \sum_k V_{fk}\,V_{ki}\left(\frac{1}{E_i - E_k} + \frac{1}{E_i + E_k}\right) = \sum_k V_{fk}\,V_{ki}\,\frac{2E_i}{E_i^2 - E_k^2}. \tag{29}$$

[4] If this troubles you, recall the Dirac-sea picture of antimatter. A plane wave $e^{iEt/\hbar}$ propagating backwards in time ($t \to -t$) becomes $e^{-iEt/\hbar}$ with negative energy, going forwards.

Up to the factor $2E_i$ (which will ultimately be absorbed into the Lorentz-invariant normalization of relativistic wavefunctions), the propagator depends on $E_i^2 - E_k^2$. With momentum conservation at each vertex ($|\vec{p}_i| = |\vec{p}_k|$), this equals $p_i^2 - p_k^2$ in terms of four-momenta, which is manifestly Lorentz invariant.
